Ryerson University

Department of Mechanical and Industrial Engineering

MTL 200: Materials Science Fundamentals

Instructor (circle): Machin | Macwan | Vandersluis

Term Test 2 - Makeup

Closed Book

March 31, 2016

Duration: 60 minutes

Name: (First Name) Maximus **(Last Name)** Makeup

Section: 75

Student Number: 03312016

Question	1	2	3	4	Total
Max. Marks	10	10	10	10	40
Marks Obtained	10	10	10	10	40

Question 1: Multiple Choice. Only one correct or most appropriate answer. Please put your answers in the table below. Only the table will be marked. **(1 mark each)**

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	D	A	D	D	B	B	A	B	B

1. Based on the phase diagram provided (page 6), how many phase fields are present?
   a) 2
   b) 3
   c) 5
   d) 6

2. A eutectic isotherm is a:
   a) Liquidus line
   b) Vertical line
   c) Solvus line
   d) Solidus line

3. $\gamma \longrightarrow \alpha + \beta$ is an example of a _____ reaction.
   a) Eutectoid
   b) Peritectic
   c) Isomorphous
   d) Eutectic

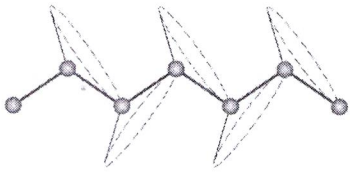
4. For an ideal cesium chloride structure, the anions are arranged as:
   a) BCC
   b) FCC
   c) HCP
   d) Simple Cubic

5. The co-ordination number for a cesium chloride structure is:
   a) 2
   b) 4
   c) 6
   d) 8

6. Ceramics are less ductile than metals because:
   a) The metal ions are smaller
   b) They have a limited number of slip systems
   c) Diffusion rates are slower
   d) The atomic packing factor is greater

7. The positioning of the carbon backbone atoms may lie anywhere on the dashed circle and still subtend an angle of _____ with the bond between the other two atoms.

   a) 90°
   b) 109°
   c) 118°
   d) 145°



8. The type of polymer which softens when heated and hardens when cooled is a:
   a) Thermoplastic polymer
   b) Copolymer
   c) Thermosetting polymer
   d) Elastomer

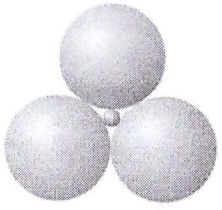
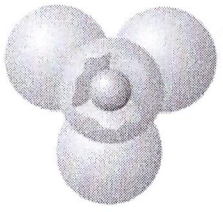
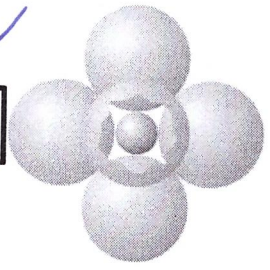
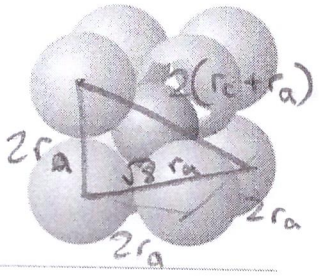
9. Highly-crystalline polyethylene will have _____ than highly-amorphous polyethylene.
   a) Lower density
   b) Lower thermal expansion
   c) Lower strength
   d) Lower melt temperature

10. Saturated hydrocarbons have:
   a) Double bonds
   b) Single bonds
   c) Triple bonds
   d) Both single and double bonds

Question 2: NiO has a rock salt structure (coordination number = 6). The ionic radii of Ni^{2+} and O^{2-} are 0.069 nm and 0.140 nm, and their atomic weights are 58.7 g/mol and 16.0 g/mol, respectively.

a) Fill in the chart below for the minimum and maximum cation-anion radius ratio for the coordination number = 6 (*for full marks, be sure to show all your work!*)

b) Find the density of NiO, given an edge length of 0.418 nm.

Coordination Number	*Cation-Anion Radius Ratio*	*Coordination Geometry*
2	< 0.155	
3	0.155–0.225	
4	0.225–0.414	
6	minimum: 0.414 / maximum: 0.732	
8	0.732 –1.0	

(a)

$2(r_c + r_a)$

$2r_a$ $\sqrt{3}\, r_a$ $2r_a$ $2r_a$

$(2r_a)^2 + (2r_a)^2 = 8r_a^2$

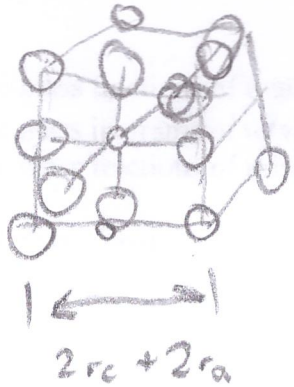
2

$8r_a^2 + 4r_a^2 = 4(r_c + r_a)^2$

$3r_a^2 = (r_c + r_a)^2$

$\sqrt{3}\, r_a = r_c + r_a \rightarrow \frac{r_c}{r_a} = (\sqrt{3} - 1)$

(b) NiO :



$a = 0.418$ nm

$r_{Ni^{2+}} = 0.069$ nm

$r_{O^{2-}} = 0.140$ nm

$A_{Ni} = 58.7$ g/mol

$A_O = 16.0$ g/mol

$$\rho = \frac{n'(\Sigma A_c + \Sigma A_a)}{V_c N_a}$$

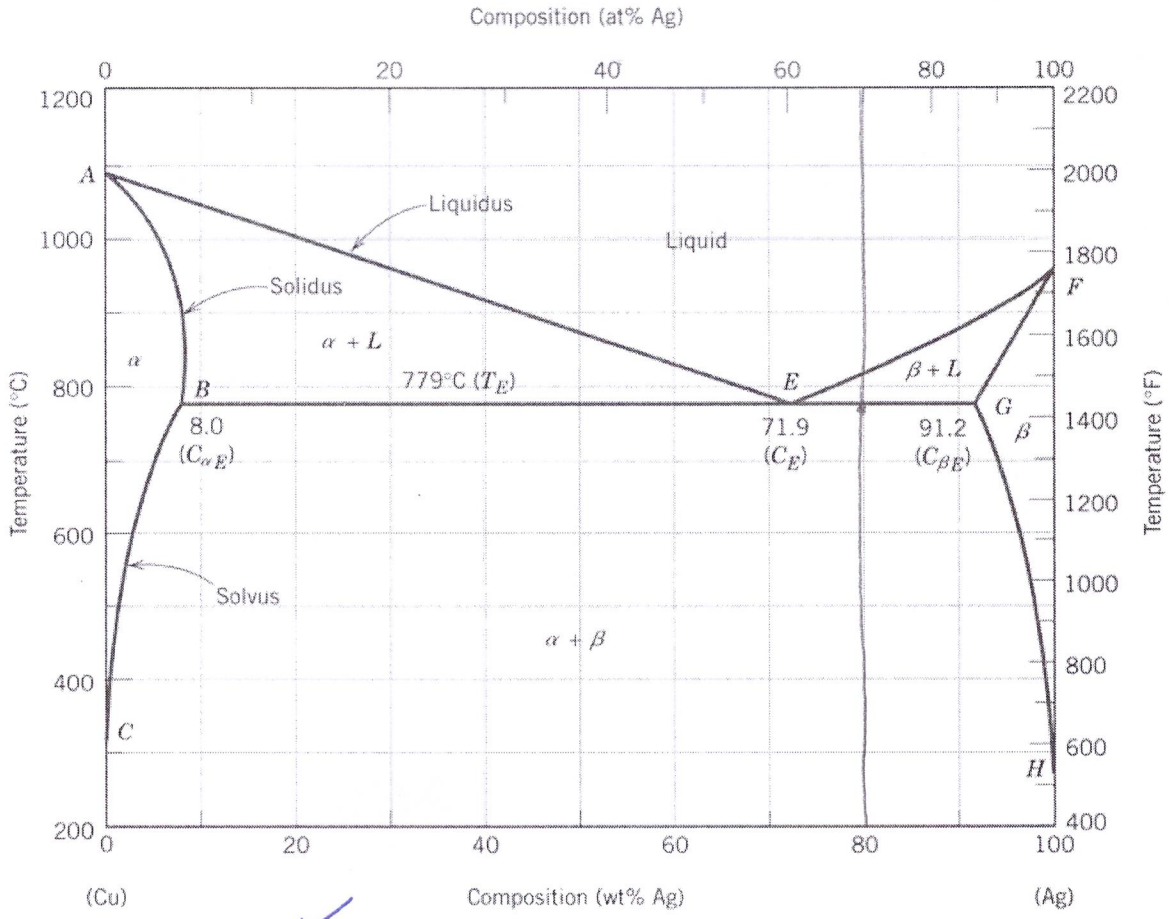
$$= \frac{(4)(58.7 + 16.0 \text{ g/mol})}{(0.418 \text{ nm})^3 (6.022 \times 10^{23} \frac{\text{units}}{\text{mol}})}$$

$$= 6.80 \times 10^{-21} \text{ g/nm}^3 \times \left(\frac{10^9 \text{ nm}}{10^2 \text{ cm}}\right)^3$$

$$= 6.80 \text{ g/cm}^3$$

Question 3: If 150 g of 80 wt% Ag – 20 wt% Cu alloy is slowly cooled from 1000 °C to just below 779 °C;

c) Determine the mass fractions of total α and total β, just below the eutectic temperature.
d) Determine the mass in grams of total β, just below the eutectic temperature.
e) Determine the mass fractions of primary α and primary β, just above the eutectic temperature.
f) Determine the mass fractions of eutectic α and eutectic β.



(a) $W_{\alpha_T} = \frac{91.2 - 80}{91.2 - 8.0} = 0.135$ ✓ ✓ $W_{\beta_T} = \frac{80 - 8.0}{91.2 - 8.0} = 0.865$ ✓

(b) $m_{\beta_T} = (0.865)(150\ g) = 130\ g$ ✓

(c) $W_{\alpha'} = 0$ ✓ $W_{\beta'} = \frac{80 - 71.9}{91.2 - 71.9} = 0.420$ ✓

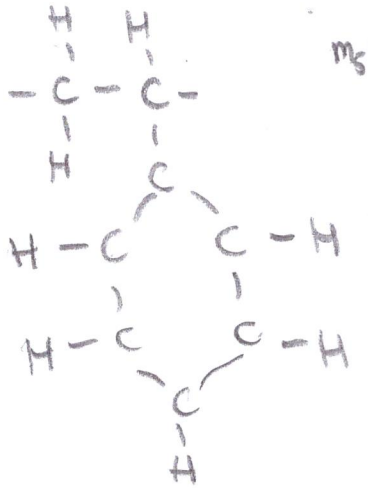
(d) $W_{\alpha_e} = W_{\alpha_T} = 0.135$ ✓ $W_{\beta_e} = W_{\beta_T} - W_{\beta'} = 0.445$ ✓

Question 4: A random poly (styrene-butadiene) copolymer has a number-average molecular weight of 350, 000 g/mol and a degree of polymerization of 5000. Compute the fraction of styrene and butadiene repeat units in this copolymer. *For full marks, be sure to show all your work!*

$$DP = \frac{\bar{M}_n}{\bar{m}}$$

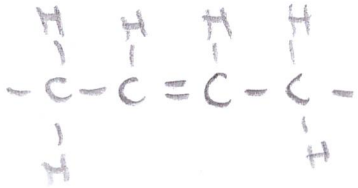
$$5000 = \frac{350\,000}{\bar{m}} \Rightarrow \bar{m} = 70 \text{ g/mol}$$

PolyStyrene :



$m_s = 8 \times 12 + 8 = 104$ g/mol

Poly butadiene :



$m_b = 4 \times 12 + 6 = 54$ g/mol

$$\bar{m} = f_s m_s + f_b m_b = f_s(104 \text{ g/mol}) + (1 - f_s)(54 \text{ g/mol}) = 70 \text{ g}$$

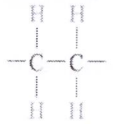
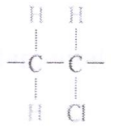
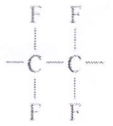
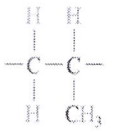
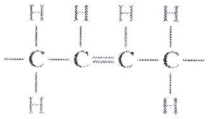
$$104 f_s + 54 - 54 f_s = 70$$

$$50 f_s = 16$$

$$f_s = 0.32 \quad \therefore f_b = 0.68$$

Formula Sheet

Table 14.3 A Listing of Repeat Units for 10 of the More Common Polymeric Materials

Polymer	*Repeat Unit*
Polyethylene (PE)	
Poly(vinyl chloride) (PVC)	
Polytetrafluoroethylene (PTFE)	
Polypropylene (PP)	
Polystyrene (PS)	
Butadiene	

Atomic Weights of Select Elements (amu)

H	C	N	O	F	S	Cl	Ni
1.0	12.0	14.0	16.0	19.0	32.1	35.5	58.7

$$DP = \frac{\overline{M_n}}{\overline{m}} \qquad \overline{m} = \Sigma\, f_j m_j$$

$$\%\ crystallinity = \frac{\rho_c\,(\rho_s - \rho_a)}{\rho_s\,(\rho_c - \rho_a)} \times 100 \qquad \rho = \frac{n'\,(\Sigma A_C + \Sigma A_A)}{V_C N_A}$$